

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : JAVA PROGRAMMING

Practical No : 6 and 7

Aim 6 : Write a java application for encoding and decoding in JSON in java

Aim 7 : JDBC (Java Database Connectivity) programs demonstrate how to interact with a database by

displaying data, fetching specific records, and performing CRUD operations (Create, Read, Update, Delete).

CODE 6:

```
package omwala_practical6;
import org.json.JSONObject;

public class Omwala_practical6 {
    static class person{
        String name;
        int age;
        String email;

        person(String name, int age, String email){
            this.name = name;
            this.age = age;
            this.email = email;
        }
    }

    public static void main(String[] args) {
        person person = new person("Om Wala", 18, "om@gmail.com");

        JSONObject jsonObject = new JSONObject();
        jsonObject.put("name", person.name);
        jsonObject.put("age", person.age);
        jsonObject.put("email", person.email);

        System.out.println("Om Wala S119");
        System.out.println("Encoded Json : ");
        System.out.println(jsonObject.toString(2));

        String jsonString = jsonObject.toString();
        JSONObject parsed = new JSONObject(jsonString);

        String name = parsed.getString("name");
        int age = parsed.getInt("age");
        String email = parsed.getString("email");

        person decodedperson = new person(name , age , email);
        System.out.println("\nDecoded Person Object : ");
        System.out.println("Name : " + decodedperson.name);
        System.out.println("Age : " + decodedperson.age);
    }
}
```

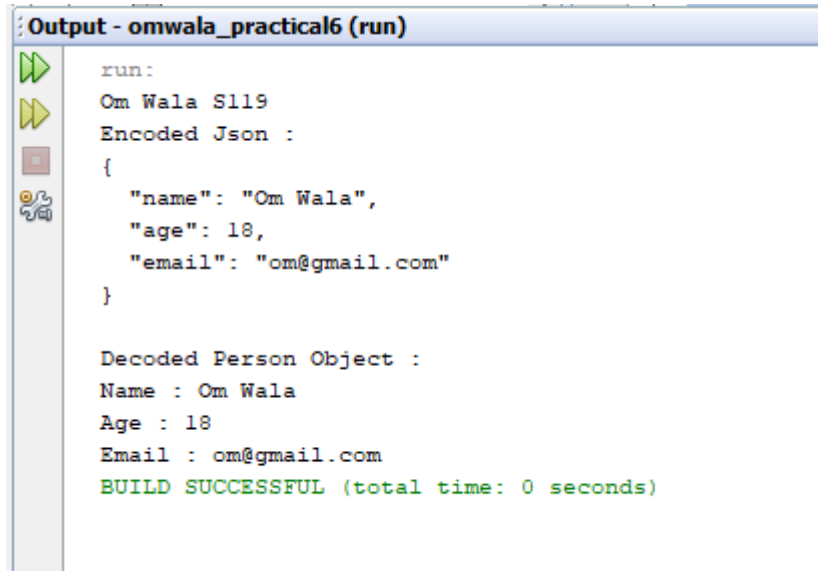
Name : Om Wala

Roll No : S119

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```
        System.out.println("Email : " + decodedperson.email);
    }
}
```

OUTPUT :



```
Output - omwala_practical6 (run)
run:
Om Wala S119
Encoded Json :
{
  "name": "Om Wala",
  "age": 18,
  "email": "om@gmail.com"
}

Decoded Person Object :
Name : Om Wala
Age : 18
Email : om@gmail.com
BUILD SUCCESSFUL (total time: 0 seconds)
```

CODE 7A:

```
package omwala_practical6;
import java.sql.*;

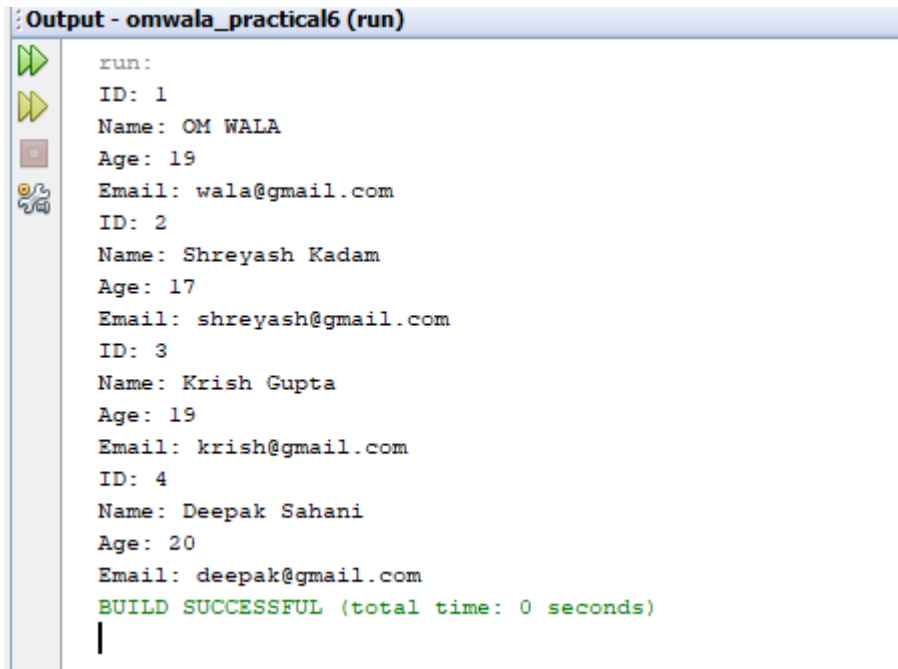
public class Omwala_practical7A {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/test_db";
        String user = "root";
        String password = "";

        String query = "SELECT * FROM employees";
        try (Connection conn = DriverManager.getConnection(url, user, password);
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(query)){
            while (rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
                int age = rs.getInt("age");
                String department = rs.getString("email");
                System.out.println("ID: " + id + "\nName: " + name + "\nAge: " + age + "\nEmail: " +
                department);
            }
        } catch (SQLException e) {e.printStackTrace();
        }
    }
}

Name : Om Wala
Roll No : S119
```

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OUTPUT :



```
run:
ID: 1
Name: OM WALA
Age: 19
Email: wala@gmail.com
ID: 2
Name: Shreyash Kadam
Age: 17
Email: shreyash@gmail.com
ID: 3
Name: Krish Gupta
Age: 19
Email: krish@gmail.com
ID: 4
Name: Deepak Sahani
Age: 20
Email: deepak@gmail.com
BUILD SUCCESSFUL (total time: 0 seconds)
```

CODE 7B:

```
package omwala_practical6;
import java.sql.*;
import java.util.Scanner;

public class Omwala_practical7B {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/test_db";
        String user = "root";
        String password = "";

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter employee ID to fetch: ");
        int employeeId = scanner.nextInt();

        String query = "SELECT * FROM employees WHERE id = ?";

        try (
            Connection conn = DriverManager.getConnection(url, user, password);
            PreparedStatement stmt = conn.prepareStatement(query)
```

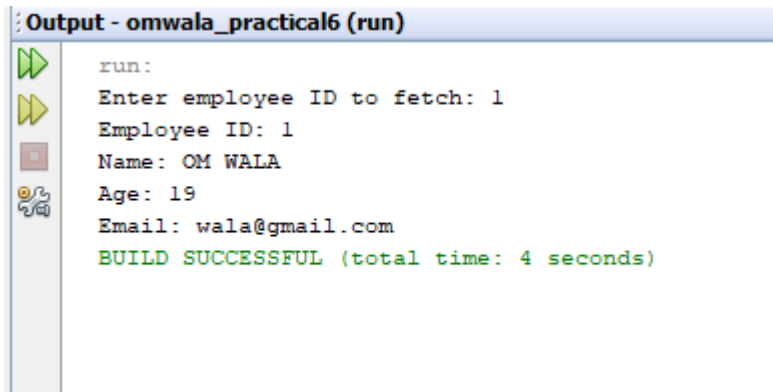
Name : Om Wala
Roll No : S119

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```
) {  
  
    stmt.setInt(1, employeeId);  
  
    try (ResultSet rs = stmt.executeQuery()) {  
  
        if (rs.next()) {  
  
            int id = rs.getInt("id");  
            String name = rs.getString("name");  
            int age = rs.getInt("age");  
            String department = rs.getString("email");  
  
            System.out.println("Employee ID: " + id);  
            System.out.println("Name: " + name);  
            System.out.println("Age: " + age);  
            System.out.println("Email: " + department);  
  
        } else {  
  
            System.out.println(  
                "No employee found with ID " + employeeId  
            );  
        }  
    }  
} catch (SQLException e) {  
  
    e.printStackTrace();  
}  
  
scanner.close();  
}  
}
```

OUTPUT :

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```
Output - omwala_practical6 (run)
run:
Enter employee ID to fetch: 1
Employee ID: 1
Name: OM WALA
Age: 19
Email: wala@gmail.com
BUILD SUCCESSFUL (total time: 4 seconds)
```

CODE 7C:

```
package omwala_practical6;
import java.sql.*;
import java.util.Scanner;
```

```
public class Omwala_practical7C {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/test_db";
        String user = "root";
        String password = "";

        Scanner scanner = new Scanner(System.in);

        System.out.println("Choose an operation: ");
        System.out.println("1. Insert");
        System.out.println("2. Update");
        System.out.println("3. Delete");

        int choice = scanner.nextInt();

        switch (choice) {

            case 1:
                insertRecord(url, user, password);
                break;

            case 2:
                updateRecord(url, user, password);
                break;

            case 3:
                deleteRecord(url, user, password);
```

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```
        break;

    default:
        System.out.println("Invalid choice");
        break;
    }
}

private static void insertRecord(String url, String user, String password) {

    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter employee ID: ");
    int id = scanner.nextInt();

    scanner.nextLine();

    System.out.print("Enter employee name: ");
    String name = scanner.nextLine();

    System.out.print("Enter employee age: ");
    int age = scanner.nextInt();

    scanner.nextLine();

    System.out.print("Enter employee email: ");
    String department = scanner.nextLine();

    String query = "INSERT INTO employees (id, name, age, email) VALUES (?, ?, ?, ?)";

    try (
        Connection conn = DriverManager.getConnection(url, user, password);
        PreparedStatement stmt = conn.prepareStatement(query)
    ) {

        stmt.setInt(1, id);
        stmt.setString(2, name);
        stmt.setInt(3, age);
        stmt.setString(4, department);

        int rowsAffected = stmt.executeUpdate();

        System.out.println(rowsAffected + " row(s) inserted.");

    } catch (SQLException e) {
        e.printStackTrace();
    }
}
```

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```
}
```

```
private static void updateRecord(String url, String user, String password) {

    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter employee ID to update: ");
    int id = scanner.nextInt();

    scanner.nextLine();

    System.out.print("Enter new employee name: ");
    String name = scanner.nextLine();

    System.out.print("Enter new employee age: ");
    int age = scanner.nextInt();

    scanner.nextLine();

    System.out.print("Enter new employee email: ");
    String department = scanner.nextLine();

    String query = "UPDATE employees SET name = ?, age = ?, email = ? WHERE id = ?";

    try (
        Connection conn = DriverManager.getConnection(url, user, password);
        PreparedStatement stmt = conn.prepareStatement(query)
    ) {

        stmt.setString(1, name);
        stmt.setInt(2, age);
        stmt.setString(3, department);
        stmt.setInt(4, id);

        int rowsAffected = stmt.executeUpdate();

        System.out.println(rowsAffected + " row(s) updated.");

    } catch (SQLException e) {
        e.printStackTrace();
    }
}
```

```
private static void deleteRecord(String url, String user, String password) {

    Scanner scanner = new Scanner(System.in);
```

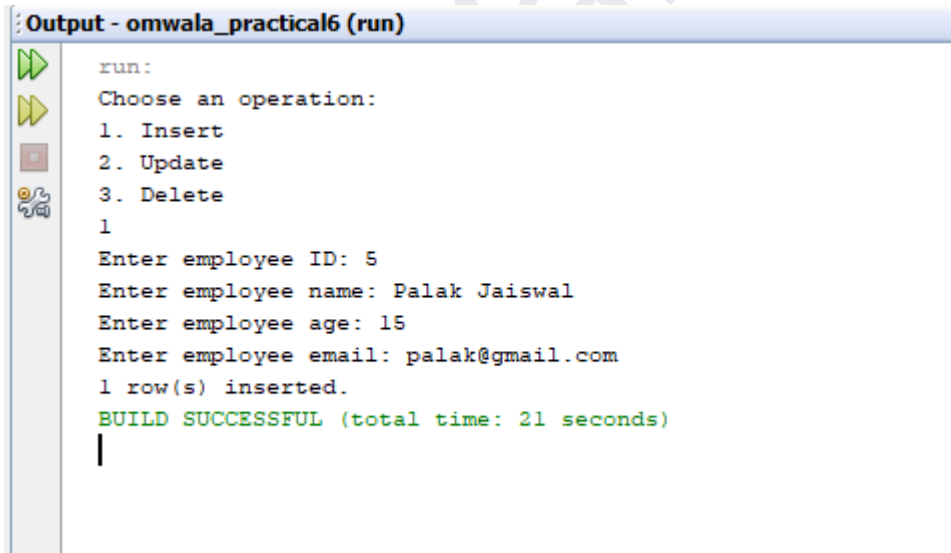
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```
System.out.print("Enter employee ID to delete: ");  
int id = scanner.nextInt();
```

```
String query = "DELETE FROM employees WHERE id = ?";
```

```
try (  
    Connection conn = DriverManager.getConnection(url, user, password);  
    PreparedStatement stmt = conn.prepareStatement(query)  
) {  
  
    stmt.setInt(1, id);  
  
    int rowsAffected = stmt.executeUpdate();  
  
    System.out.println(rowsAffected + " row(s) deleted.");  
  
} catch (SQLException e) {  
    e.printStackTrace();  
}  
}
```

OUTPUT :



```
run:  
Choose an operation:  
1. Insert  
2. Update  
3. Delete  
1  
Enter employee ID: 5  
Enter employee name: Palak Jaiswal  
Enter employee age: 15  
Enter employee email: palak@gmail.com  
1 row(s) inserted.  
BUILD SUCCESSFUL (total time: 21 seconds)
```

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```
Output - omwala_practical6 (run)

run:
Choose an operation:
1. Insert
2. Update
3. Delete
3
Enter employee ID to delete: 5
1 row(s) deleted.
BUILD SUCCESSFUL (total time: 49 seconds)
```

```
mysql> select * from employees;
+----+-----+-----+-----+
| id | name          | age | email                |
+----+-----+-----+-----+
| 1  | Om Wala       | 18  | om@gmail.com         |
| 2  | Shreyash Kadam | 17  | shreyash@gmail.com   |
| 3  | Krish Gupta   | 19  | krish@gmail.com      |
| 4  | Deepak Sahani | 20  | deepak@gmail.com     |
| 5  | Palak Jaiswal | 15  | palak@gmail.com      |
+----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> select * from employees;
+----+-----+-----+-----+
| id | name          | age | email                |
+----+-----+-----+-----+
| 1  | Om Wala       | 18  | om@gmail.com         |
| 2  | Shreyash Kadam | 17  | shreyash@gmail.com   |
| 3  | Krish Gupta   | 19  | krish@gmail.com      |
| 4  | Deepak Sahani | 20  | deepak@gmail.com     |
+----+-----+-----+-----+
4 rows in set (0.00 sec)
```

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Output - omwala_practical6 (run)

```
run:
Choose an operation:
1. Insert
2. Update
3. Delete
2
Enter employee ID to update: 1
Enter new employee name: OM WALA
Enter new employee age: 19
Enter new employee email: wala@gmail.com
1 row(s) updated.
BUILD SUCCESSFUL (total time: 24 seconds)
```

```
mysql> select * from employees;
```

id	name	age	email
1	OM WALA	19	wala@gmail.com
2	Shreyash Kadam	17	shreyash@gmail.com
3	Krish Gupta	19	krish@gmail.com
4	Deepak Sahani	20	deepak@gmail.com

4 rows in set (0.00 sec)